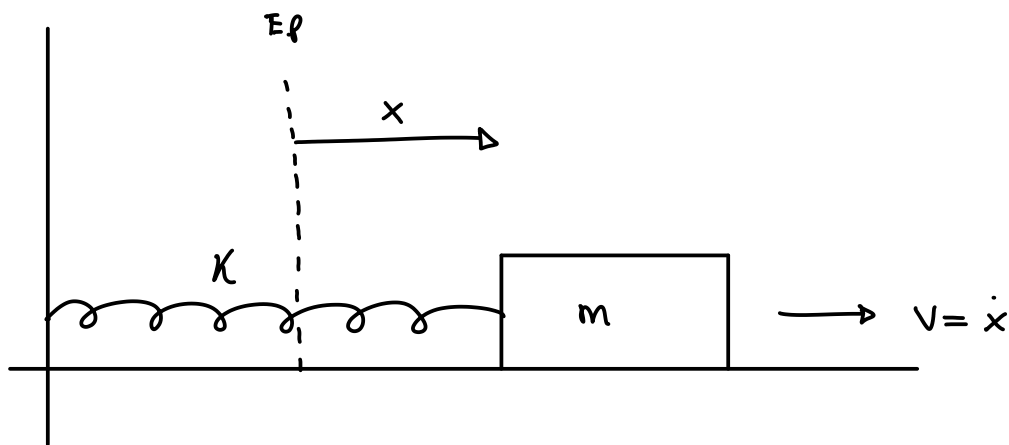




$$T = \frac{1}{2} m \dot{x}^2$$

$$U = \frac{1}{2} k x^2$$

$$L = T - U \Rightarrow L = \frac{1}{2} m \dot{x}^2 - \frac{1}{2} k x^2$$

$$\frac{d}{dt} \left(\frac{\partial L}{\partial \dot{x}} \right) - \frac{\partial L}{\partial x} = 0 \Rightarrow \frac{d}{dt} (m \dot{x}) + k x = 0$$

$$\Rightarrow m \ddot{x} = -k x$$

$$\ddot{x} = -\frac{k}{m} x$$

Propuesta de sol.

$$x(t) = A \cos(\omega t)$$

$$\dot{x}(t) = -A \omega \sin(\omega t)$$

$$\ddot{x}(t) = -A \omega^2 \cos(\omega t)$$

Sustituyendo

$$\ddot{x}(t) = -\frac{k}{m} x \Rightarrow -\cancel{A \omega^2 \cos(\omega t)} = -\frac{k}{m} A \cancel{\cos(\omega t)}$$

$$\ddot{x}(t) = \omega^2 x$$

$\omega^2 =$ Frecuencia angular.

$$-\omega^2 = -\frac{k}{m}$$

$$\omega^2 = \frac{k}{m}$$

Lagrangian for a charge in magnetic field

Hallar:

$$m\ddot{\vec{r}} = q(\vec{E} + \dot{\vec{r}} \times \vec{B}) \quad \leftarrow \text{Fuerza de Lorentz}$$

Maxwell equations ...

$$\vec{\nabla} \cdot \vec{E} = \frac{\rho_0}{\epsilon_0} \quad \text{Gauss}$$

$$\vec{\nabla} \times \vec{E} = -\frac{\partial \vec{B}}{\partial t} \quad \text{Faraday}$$

$$\vec{\nabla} \cdot \vec{B} = 0 \quad \text{Gauss}$$

$$\vec{\nabla} \times \vec{B} = \mu_0 \vec{J} + \mu_0 \epsilon_0 \frac{\partial \vec{E}}{\partial t} \quad \text{Ampere - Maxwell}$$

In electrostatics

$$\vec{\nabla} \times \vec{E} = 0 \Rightarrow \vec{E} = -\nabla V$$

$$\vec{\nabla} \cdot \vec{B} = 0$$

curl of a grad is zero, div of a curl is zero.

$$\vec{B} = \vec{\nabla} \times \vec{A} \quad \leftarrow \text{potential vectorial}$$

$$\vec{\nabla} \times \vec{E} = -\frac{\partial}{\partial t} (\vec{\nabla} \times \vec{A})$$

curl of a grad is zero

$$\Rightarrow \vec{\nabla} \times \left(\vec{E} + \frac{\partial}{\partial t} \vec{A} \right) = 0$$

$$\vec{E} + \frac{\partial}{\partial t} \vec{A} = -\nabla V$$

Def.

$$\left. \begin{aligned} \vec{E} &= -\nabla V - \frac{\partial}{\partial t} \vec{A} \\ \vec{B} &= \vec{\nabla} \times \vec{A} \end{aligned} \right\}$$

Now we can write the Lagrangian

Consider the following Lagrangian:

$$\begin{aligned}
 L(\vec{r}, \dot{\vec{r}}, t) &= \frac{1}{2} m \dot{\vec{r}}^2 - q \left(U - \dot{\vec{r}} \cdot \vec{A} \right) \\
 &\quad \uparrow \text{potential } U(x, y, z, t) \\
 &\quad \quad \quad \vec{A}(x, y, z, t) \\
 &= \frac{1}{2} m (\dot{x}^2 + \dot{y}^2 + \dot{z}^2) - q \left(U - \dot{x} A_x - \dot{y} A_y - \dot{z} A_z \right)
 \end{aligned}$$

One component $\rightarrow$ Lorentz equation

$$\frac{\partial L}{\partial x} = -q \left(\frac{\partial U}{\partial x} - \dot{x} \frac{\partial A_x}{\partial x} - \dot{y} \frac{\partial A_y}{\partial x} - \dot{z} \frac{\partial A_z}{\partial x} \right)$$

$$\begin{aligned}
 \frac{\partial L}{\partial \dot{x}} &= m \dot{x} + q A_x \quad ; \quad \frac{d}{dt} \left(\frac{\partial L}{\partial \dot{x}} \right) = m \ddot{x} + q \left(\frac{\partial A_x}{\partial x} \dot{x} + \frac{\partial A_x}{\partial y} \dot{y} + \frac{\partial A_x}{\partial z} \dot{z} + \frac{\partial A_x}{\partial t} \right) \\
 &\quad \quad \quad // \text{chain rule for } \frac{d}{dt} A_x(x, y, z, t)
 \end{aligned}$$

$$\begin{aligned}
 \frac{d}{dt} \left(\frac{\partial L}{\partial \dot{x}} \right) - \frac{\partial L}{\partial x} &= 0 \quad \left\{ m \ddot{x} + q \left(\frac{\partial A_x}{\partial x} \dot{x} + \frac{\partial A_x}{\partial y} \dot{y} + \frac{\partial A_x}{\partial z} \dot{z} + \frac{\partial A_x}{\partial t} \right) \right. \\
 &\quad \left. - \left[-q \left(\frac{\partial U}{\partial x} - \frac{\partial A_x}{\partial x} \dot{x} - \frac{\partial A_y}{\partial x} \dot{y} - \frac{\partial A_z}{\partial x} \dot{z} \right) \right] \right\} = 0
 \end{aligned}$$

$$\Rightarrow m \ddot{x} + q \left(\frac{\partial A_x}{\partial x} \dot{x} + \frac{\partial A_x}{\partial y} \dot{y} + \frac{\partial A_x}{\partial z} \dot{z} + \frac{\partial A_x}{\partial t} \right) + q \left(\frac{\partial U}{\partial x} - \frac{\partial A_x}{\partial x} \dot{x} - \frac{\partial A_y}{\partial x} \dot{y} - \frac{\partial A_z}{\partial x} \dot{z} \right) = 0$$

$$\Rightarrow m \ddot{x} = -q \left(\frac{\partial U}{\partial x} + \frac{\partial A_x}{\partial t} \right) + q \dot{y} \left(\frac{\partial A_y}{\partial x} - \frac{\partial A_x}{\partial y} \right) - q \dot{z} \left(\frac{\partial A_x}{\partial z} - \frac{\partial A_z}{\partial x} \right)$$

$-E_x$ B_z B_y

$$\vec{E} = -\nabla V - \frac{\partial}{\partial t} \vec{A}$$

$$\vec{B} = \vec{\nabla} \times \vec{A} = \begin{vmatrix} \hat{x} & \hat{y} & \hat{z} \\ \partial_x & \partial_y & \partial_z \\ A_x & A_y & A_z \end{vmatrix} = (\partial_y A_z - \partial_z A_y) \hat{x} - (\partial_x A_z - \partial_z A_x) \hat{y} + (\partial_x A_y - \partial_y A_x) \hat{z}$$

$$\vec{v} \times \vec{B} = \begin{vmatrix} \hat{x} & \hat{y} & \hat{z} \\ v_x & v_y & v_z \\ B_x & B_y & B_z \end{vmatrix}; \quad (\vec{v} \times \vec{B})_x = v_y B_z - v_z B_y$$

$$\Rightarrow m \ddot{x} = q(E_x + (\vec{v} \times \vec{B})_x) \quad \leftarrow \text{Correct E.O.M}$$

$$L = \frac{1}{2} m \dot{\vec{r}}^2 - q(V - \dot{\vec{r}} \cdot \vec{A})$$

$$p_x = \frac{\partial L}{\partial \dot{x}} = m \dot{x} + q A_x$$

Canonical momentum

$$\vec{p} = m \vec{v} + q \vec{A} \rightarrow \text{conserved quantity.}$$